Tarea lección 3.3

1) $f(x) = x^2 \operatorname{sen} x$

$f'(x) = x^2 \cos x + 2x \operatorname{sen} x$

5) $g(t) = t^3 \cos t$

$g'(t) = 3t^2 \cos t + t^3(-\operatorname{sen} t) = 3t^2 \cos t - t^3 \operatorname{sen} t$

9) $y = \dfrac{x}{2 - \tan x}$

$y' = \dfrac{2 - \tan x (1) - x(-\sec^2 x)}{(2 - \tan x)^2} = \dfrac{2 - \tan x + x \sec^2 x}{(2 - \tan x)^2}$

13) $y = \dfrac{t \operatorname{sen} t}{1 + t}$

$y' = \dfrac{(t \operatorname{sen} t)(1 + x) - (x \operatorname{sen} x)(1 + x)}{(1 + t)^2} = \dfrac{(\operatorname{sen} x + x \cos x)(1 + x) - (x \operatorname{sen} x)(x+1)}{(1 + t)^2}$

$y' = \dfrac{\operatorname{sen} x + x \cos x + x^2 \cos x}{(1 + x)^2}$

15) $f(\theta) = \theta \cos\theta \operatorname{sen}\theta$

$f'(\theta) = \theta \cos\theta \cos\theta + \operatorname{sen}\theta(-\operatorname{sen}\theta) = \frac{1}{2} \operatorname{sen}\theta + \theta \cos 2\theta$

19) Demuestre $\frac{d}{dx}(\cot x) = -\csc^2 x$

$\cot x = -\csc^2 x$

21) $y = \text{sen}x + \cos x \qquad (0,1)$

$y' = \cos x - \text{sen}x \qquad y'|_0 = \cos(1) - \text{sen}(1) = 1$

$y - 1 = 1(x - 0)$

$y = \underline{x + 1}$

31) a) $f(x) = \dfrac{\tan x - 1}{\sec x}$

$f'(x) = \dfrac{\sec x(\sec^2 x) - (\tan x - 1)(\sec x \tan x)}{(\sec x)^2}$

$f'(x) = \dfrac{\sec^2 x - (\tan x)(\tan x - 1)}{\sec x}$

b) $f(x) = \dfrac{\tan x - 1}{\sec x} = f'(x) = \dfrac{\frac{\text{sen}x}{\cos x} - \cos x}{\frac{1}{\cos x}}$

$f(x) = \text{sen}x - \cos x$

$f'(x) = \cos x - \text{sen}x$

$\cos x + \text{sen}x = \dfrac{1 + \tan x}{\sec x}$

$\cos x + \text{sen}x = \dfrac{1 + \frac{\text{sen}x}{\cos x}}{\frac{1}{\cos x}}$

$$\dfrac{\frac{\cos x + \text{sen}x}{\cos x}}{\frac{1}{\cos x}}$$

$\cos x + \text{sen}x = \underline{\cos x + \text{sen}x}$